

MovieLens

BIG DATA TOOLS & TECHNIQUES

Group 11

Kirichenko, Gavriel
Arul, Callum Joseph
Bhattarai, Diparshan
Chang, Chia-Wei
Giri, Bikash
Hussainyar, Friba
Liu, Hsi-Teng
Medrano Diaz, Sergio Alejandro
Omisesan, Abdul-Rasaq

Dataset Description

Data Source: MovieLens (provided by GroupLens)

This data has been cleaned up - users who had less than 20 ratings or did not have complete demographic information were removed from this data set

1. Data Format:

- The *.dat files are plain text files, with :: (double colons) as the delimiter.
- 100000 ratings by 943 users on 1682 items. Each user has rated at least 20 movies. Users and items are numbered consecutively from 1. The data is randomly ordered.

2. Structured Data:

- Movies data is display as follows: Movie_Id | Movie_Title | Release_Date | Video_Release_Date | Imdb_Url | All the genres...
- The last 19 fields are the genres, a 1 indicates the movie is of that genre, a 0 indicates it is not; movies can be in several genres at once.
- Ratings data is display as follows: UserID | Movie_Id | Rating | Timestamp
- Users data is display as follows: UserID | Gender | Age | Occupation | Zip-code

Business Questions We Aim to Answer

1. Which movies are the most popular among different age groups?
2. Do male and female users have different rating preferences?
3. Which movie genres have the highest average ratings?
4. How are users' ratings affected by movie release date?
5. How does movie rating behavior vary by user occupation?
6. Which users are most active in terms of number of ratings?

ETL Pipeline Summary

1. Extract

- We sourced the raw data from the MovieLens dataset, originally provided as `.dat` files (`u.user`, `u.item`, and `u.data`).
- These files were extracted locally using Python.

2. Transform

- We transformed the raw data into structured `.csv` files and applied the following processing steps:
 - Parsed and renamed all fields
 - Dropped unnecessary columns like `zip\_code`
 - One-hot encoded genres into binary columns
 - Merged the `users`, `ratings`, and `movies` datasets into a unified format
 - Filtered users who had less than 20 ratings
 - Ensured consistency in data types and structure for analysis
- We also created new fields such as:
 - Average rating per user group (age, gender, occupation)
 - Rating count per user
 - Movie popularity and genre-wise ratings
- Finally, we stored the cleaned datasets in MongoDB

3. Load

The cleaned data was:

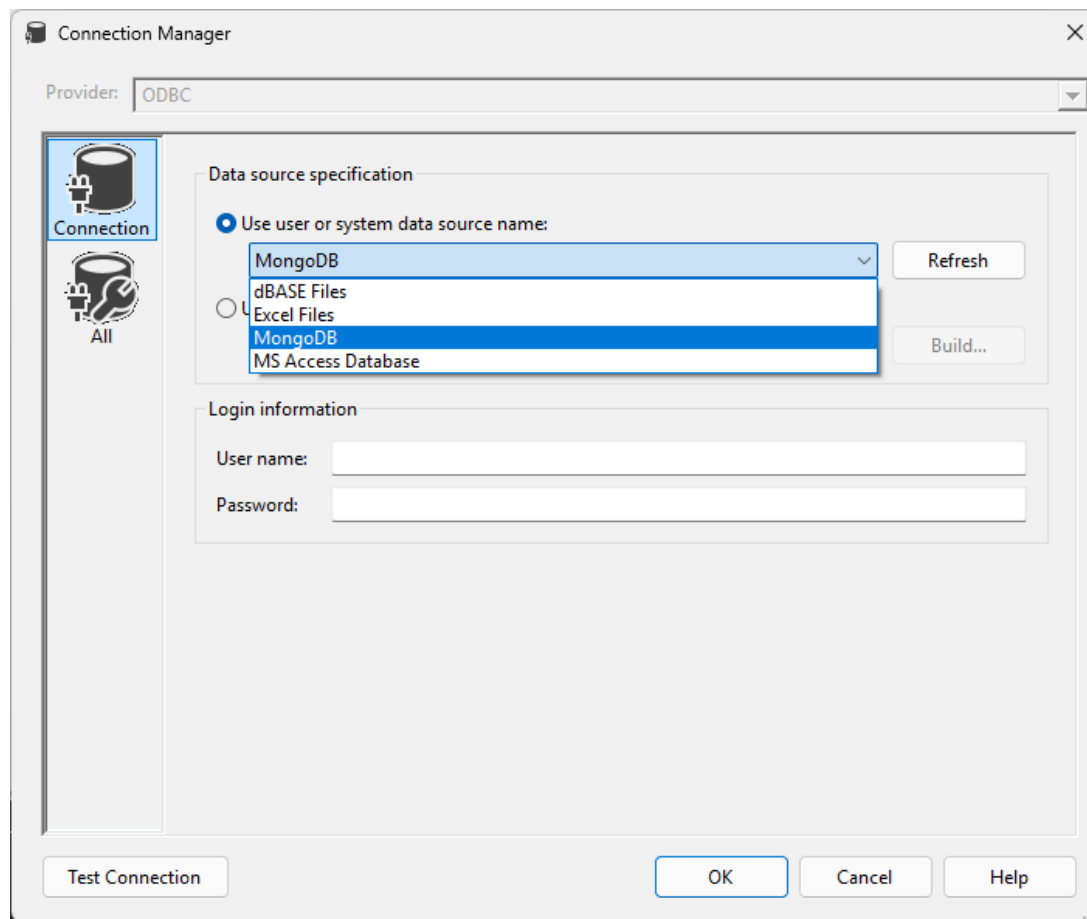
- Loaded from MongoDB
- Loaded into Pandas dataframes for further analysis and visualization
- Loaded into Power BI for interactive dashboards

This ETL pipeline allowed us to prepare high-quality, business-ready data for answering our analysis questions.

Import Data from MongoDB to MS SQL Server

First Attempt in SSIS with Mongo BI Connector

Mongo BI Connector provider a SQL daemon pairs with an ODBC driver. The import was possible, but the transform to normal forms was too difficult in SSIS.



The Mismatch Between Semi-Structured and RDBM Structured Data

```
_id: ObjectId('67e36452fb30574487df494b')
Movie_Id : 1
Movie_Title : "Toy Story (1995)"
Release_Date : "01-Jan-1995"
Video_Release_Date : NaN
Imdb_Url : "http://us.imdb.com/M/title-exac
Unknown : 0
Action : 0
Adventure : 0
Animation : 1
Children : 1
Comedy : 1
Crime : 0
Documentary : 0
Drama : 0
Fantasy : 0
Film-Noir : 0
Horror : 0
Musical : 0
```

MovieId	GenreName
1	Animation
1	Children
1	Comedy
2	Action
2	Adventure
2	Thriller
3	Thriller
4	Action
4	Comedy
4	Drama
5	Crime
5	Drama
5	Thriller
6	Drama
7	Drama

What makes it difficult to transform data in SSIS is the shapes of source and target are too different. The most significant example is that the source structure of a movie is consist of properties of genre, each property represents a genre, 1 indicates the movie belongs to the genre, 0 otherwise. When in RDBM, the normalized form is a genre table with many to one relation to the movie table. This proves difficult to transform in SSIS.

The SSIS Script Component

The script component was considered as the transformation solution. However, the object model of script component support one-to-one transform, while we need one-to-many for turning fields of possible genre to multiple records.

Furthermore, each script component is based on a full .NET Framework project which is on the verge of being outdated. The project is heavy and not worthy of being called a script

Import Using Python Script

A short python script with pyodbc can efficiently bulk insert into MS SQL Server. The queries to MongoDB Atlas is simple and fast. Transformation of data such as converting string to date is performed in the script as well.

Data Error

An invalid movie was titled “unknown” with no genre and the corresponding release date was NaN.

SSAS

A SSAS project was created. Importing data from MS SQL Server is a smooth process.

New Columns

Three columns were added to the imported data.

- Movie.ReleaseYear
Trasformed from Movie.ReleaseDate
- MovieRating.Title
The Title column is from Movie.Title
- MovieUser.AgeGroup
Calculated from MovieUser.Age

Metrics

The following are the metrics constructed inside the SSAS project.

- AllMoviesCount:= COUNT(Movie[MovieId])
- MovieReleaseYearCount:= DISTINCTCOUNT(Movie[ReleaseYear])
- GenreCount:= DISTINCTCOUNT(MovieGenre[GenreName])
- RatedMovieCount:= DISTINCTCOUNT(MovieRating[MovieId])
- UserCount:= COUNT(MovieUser[UserId])
- GenderCount:= DISTINCTCOUNT(MovieUser[Gender])
- MaleCount:= COUNTX(FILTER(MovieUser, [Gender]="M"), [Gender])
- FemaleCount:= COUNTAX(FILTER(MovieUser, [Gender]="F"), [UserId])
- OccupationCount:= DISTINCTCOUNT(MovieUser[Occupation])
- AgeGroupCount:= DISTINCTCOUNT(MovieUser[AgeGroup])
- AgeGroup0:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 0), [AgeGroup])
- AgeGroup10:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 10), [AgeGroup])
- AgeGroup20:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 20), [AgeGroup])
- AgeGroup30:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 30), [AgeGroup])
- AgeGroup40:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 40), [AgeGroup])
- AgeGroup50:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 50), [AgeGroup])
- AgeGroup60:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 60), [AgeGroup])
- AgeGroup70:= COUNTAX(FILTER(MovieUser, [AgeGroup] = 70), [AgeGroup])

PowerBI Exploration

1. User Demographics & Preferences Report

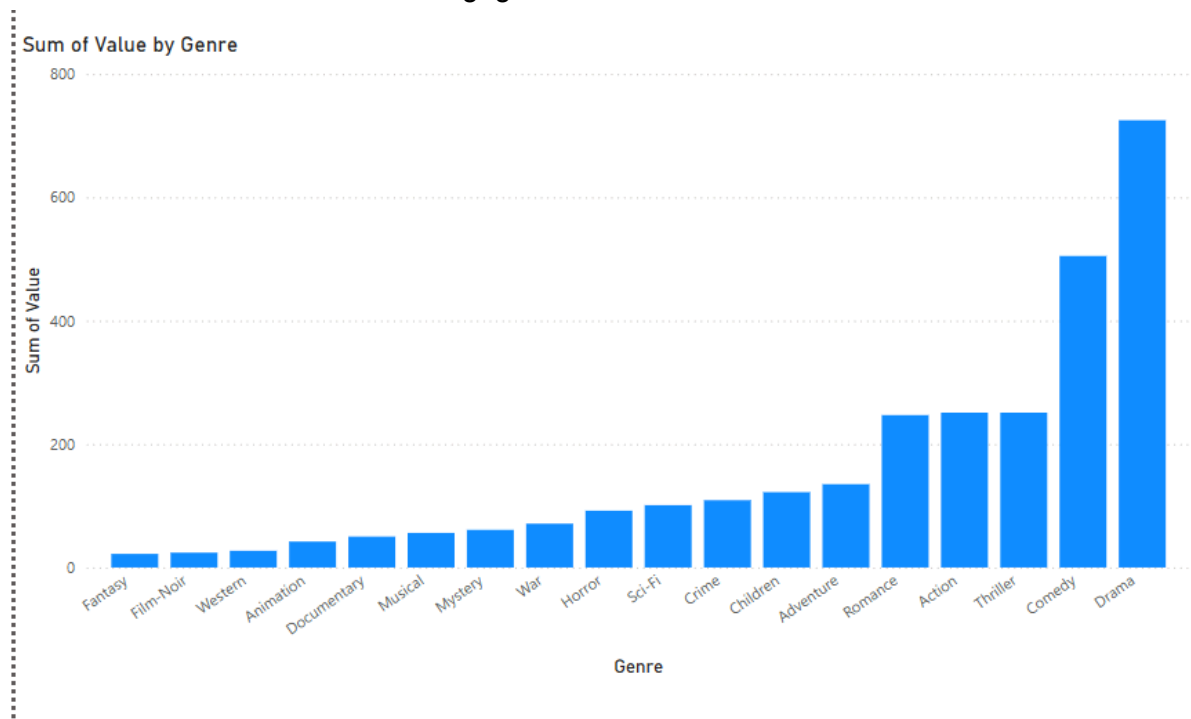
This report provides insights into the user demographics and preferences by analyzing gender, occupation, and age distribution.

Gender Distribution: A pie chart shows the proportion of male and female users.

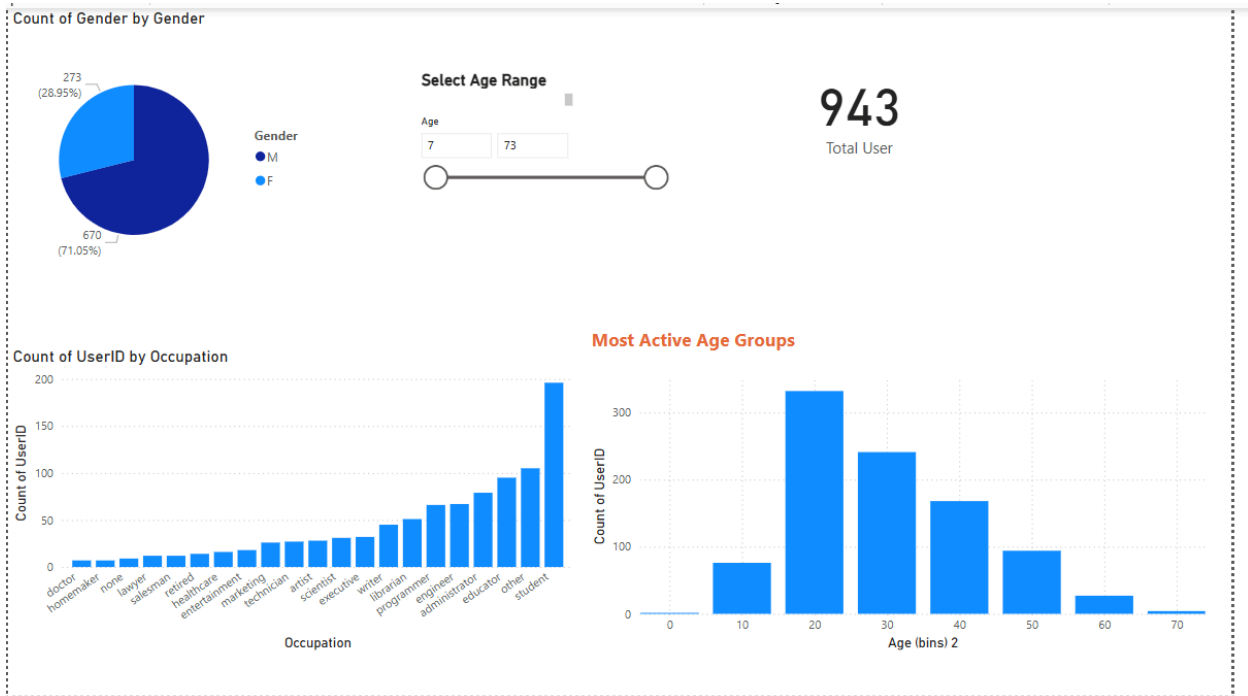
User Count by Occupation: A bar chart highlights the most common occupations among users, with students being the largest group.

Total Users: A summary card displays the total number of users (943).

Age Distribution: A bar chart identifies the most active age groups, showing that users in their 20s and 30s are the most engaged.



Age Filter: A slider allows for filtering data based on specific age ranges.

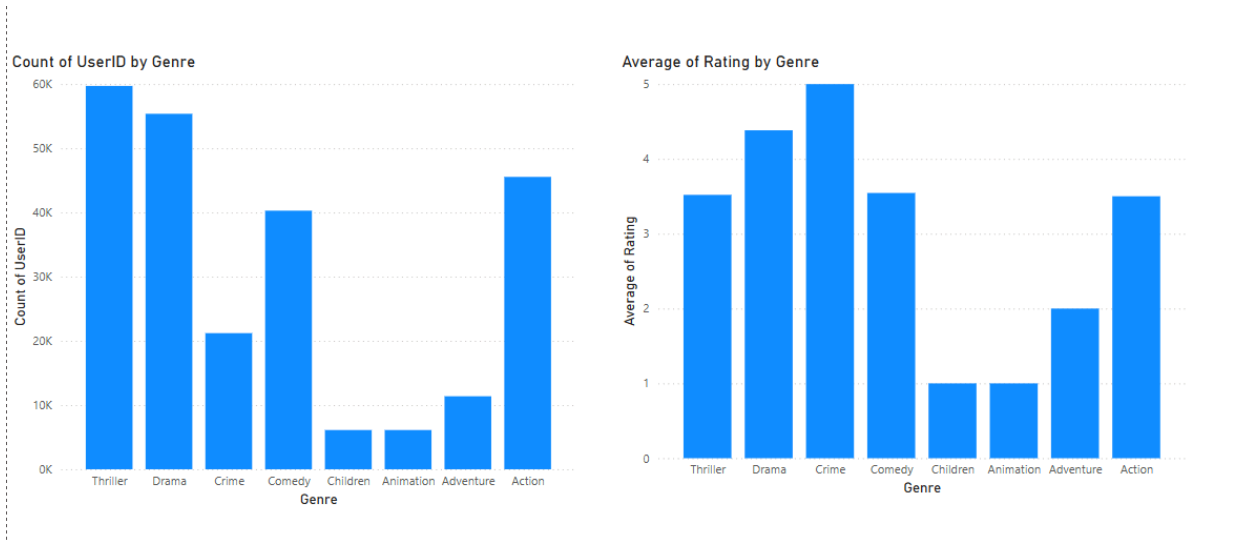


2. Movie Genre Popularity Report

This chart provides insights into movie genre popularity based on user interactions and ratings.

Left Chart (User Count by Genre): Shows how many users watched or rated movies in each genre. Thriller and Drama have the highest user engagement, while Animation and Children's genres have the lowest.

Right Chart (Average Rating by Genre): Displays the average rating per genre. Crime and Drama receive the highest ratings, while Animation and Children's genres have the lowest.



3. Top-Rated Movies Report

This Power BI report displays movie ratings insights

Table (Left): Lists movies with their **average rating**, sorted from highest to lowest.

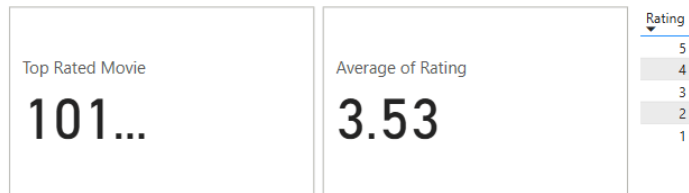
"Top Rated Movie" Card (Middle-Left): Supposed to show the highest-rated movie, but the title is truncated ("101...").

"Average of Rating" Card (Middle-Right): Displays the **overall average rating** (3.53).

Rating Filter (Right): Allows filtering movies by their rating values (1–5).

◆ **Key Insight:** "Copycat (1995)" has the highest rating (5.00). The display issue in the "Top Rated Movie"

Movie_Title	Average of Rating
Copycat (1995)	5.00
Get Shorty (1995)	4.00
Four Rooms (1995)	3.00
GoldenEye (1995)	2.00
Toy Story (1995)	1.00
Total	3.53

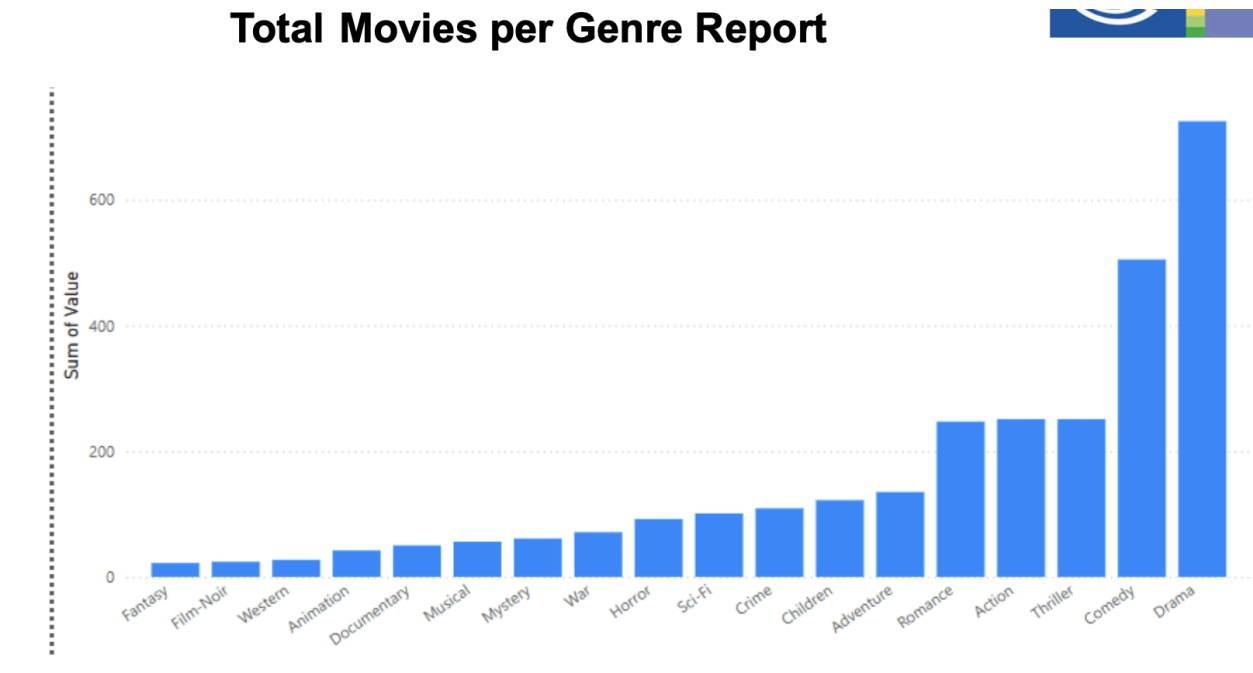


4. Total Movies per Genre Report

This bar chart displays the total number of movies for each genre in the dataset.

The **Drama** genre has the highest number of movies, followed by **Comedy** and **Action**, while genres like **Film-Noir** and **Fantasy** have fewer movies.

By analyzing this report, we can understand which genres are most prevalent in the dataset and identify trends in movie production across different categories.



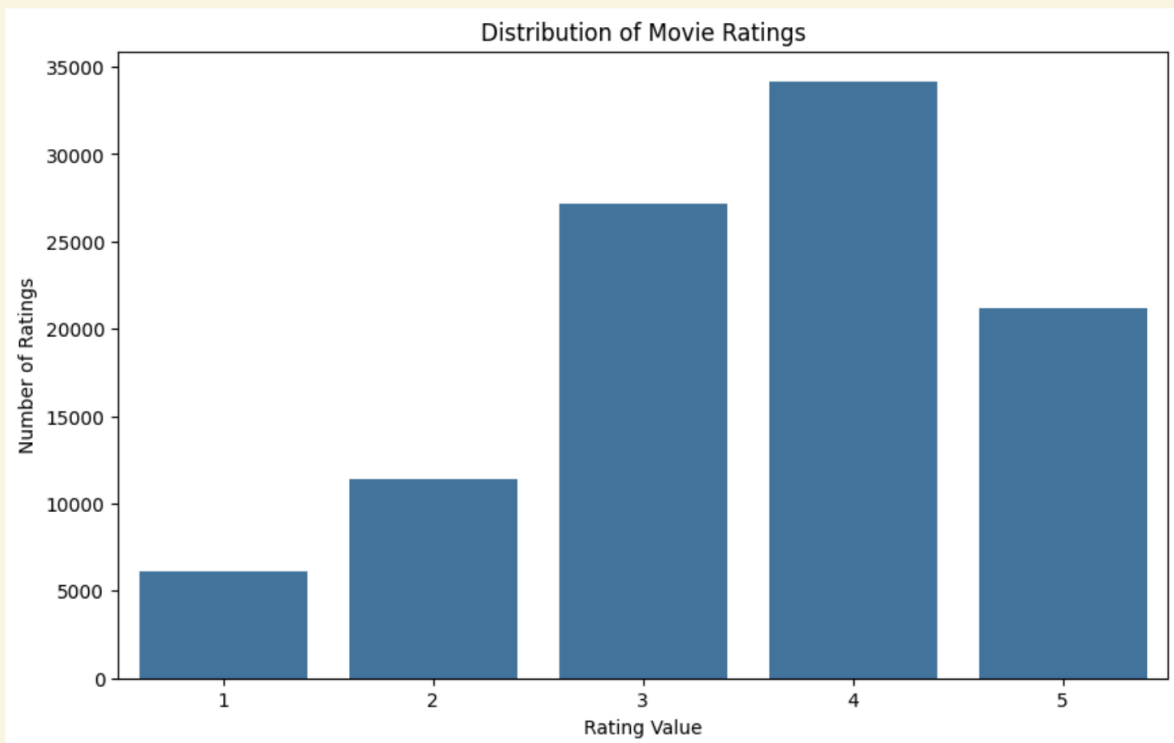
Data Analysis

Rating Distribution Analysis

Most users give ratings of 3, 4, or 5, with 4 being the most common. Lower ratings (1 and 2) are less frequent. The distribution suggests a tendency toward positive or average ratings.

✓ Rating Distribution Analysis:

	count	rating	unique_users
0	6110	1	723
1	11370	2	894
2	27145	3	936
3	34174	4	942
4	21201	5	928

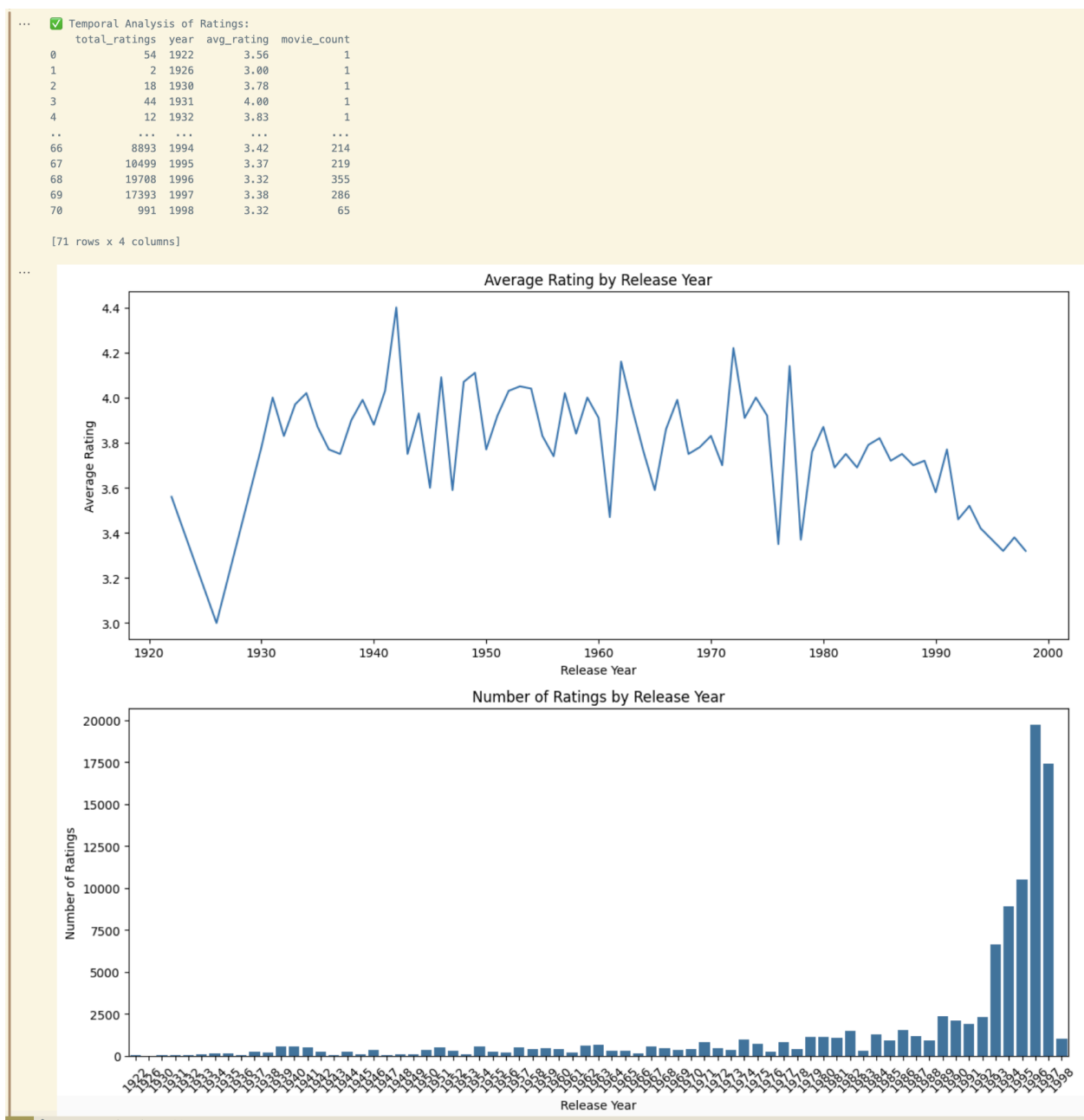


+ Code

+ Markdown

Temporal Analysis

Older movies (1930s--1960s) tend to have higher average ratings, with a noticeable decline in average ratings for more recent releases. This may reflect nostalgia bias or the longevity of classic films. The number of ratings increases sharply for movies released after 1980, peaking in the 1990s. This trend suggests users engage more with newer films, possibly due to greater availability and popularity.



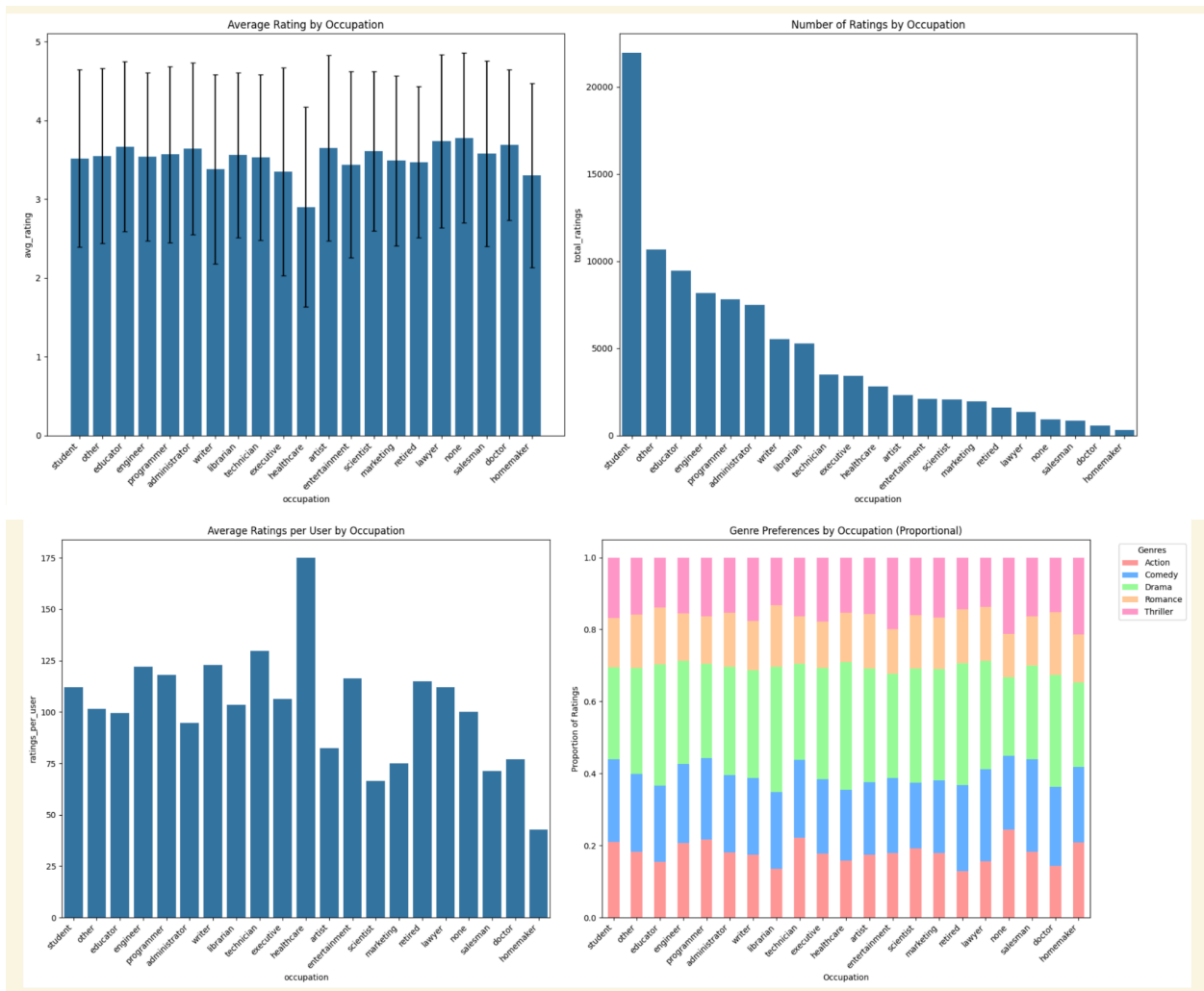
Gender-Based Movie Preferences Analysis

Female users tend to rate genres like Musical, Children, and Romance higher than male users, with the largest difference seen in Musicals (+0.17). Males gave slightly higher ratings for genres like Western, Mystery, and Sci-Fi. Despite these differences, overall rating trends across genders are relatively similar. This suggests modest but notable preference variation by gender across certain genres.



Occupation-Based Analysis

Students are the most active in terms of total ratings, while healthcare professionals rate the most per user. Average ratings are fairly consistent across occupations, though artists and homemakers rate slightly lower. Genre preferences are broadly similar, but some variation exists---e.g., administrators lean toward drama, and executives favor comedy. Standard deviations indicate that most groups show wide variation in ratings, with entertainment and writing-related occupations being the most consistent raters.



...
Most active occupations (by ratings per user):

	occupation	ratings_per_user
10	healthcare	175.2
8	technician	129.8
6	writer	123.0
3	engineer	122.0
4	programmer	118.2
12	entertainment	116.3
15	retired	114.9
16	lawyer	112.1
0	student	112.0
9	executive	106.3
7	librarian	103.4
1	other	101.5
17	none	100.0
2	educator	99.4
5	administrator	94.7
11	artist	82.4
19	doctor	77.1
14	marketing	75.0
18	salesman	71.3
13	scientist	66.4
20	homemaker	42.7

...

12	entertainment	1.18
6	writer	1.20
10	healthcare	1.27
9	executive	1.32

Rating Trends Analysis

The most rated movies include well-known titles like ‘Star Wars’ and ‘Return of the Jedi’, both with high average ratings above 4.0. The scatter plot shows that movies with more ratings tend to have higher average ratings, though there's a wide spread among less-rated films. This suggests that popular movies may benefit from rating bias due to fan enthusiasm or cultural impact. Meanwhile, movies with fewer ratings exhibit more variability, possibly due to niche appeal or limited exposure.

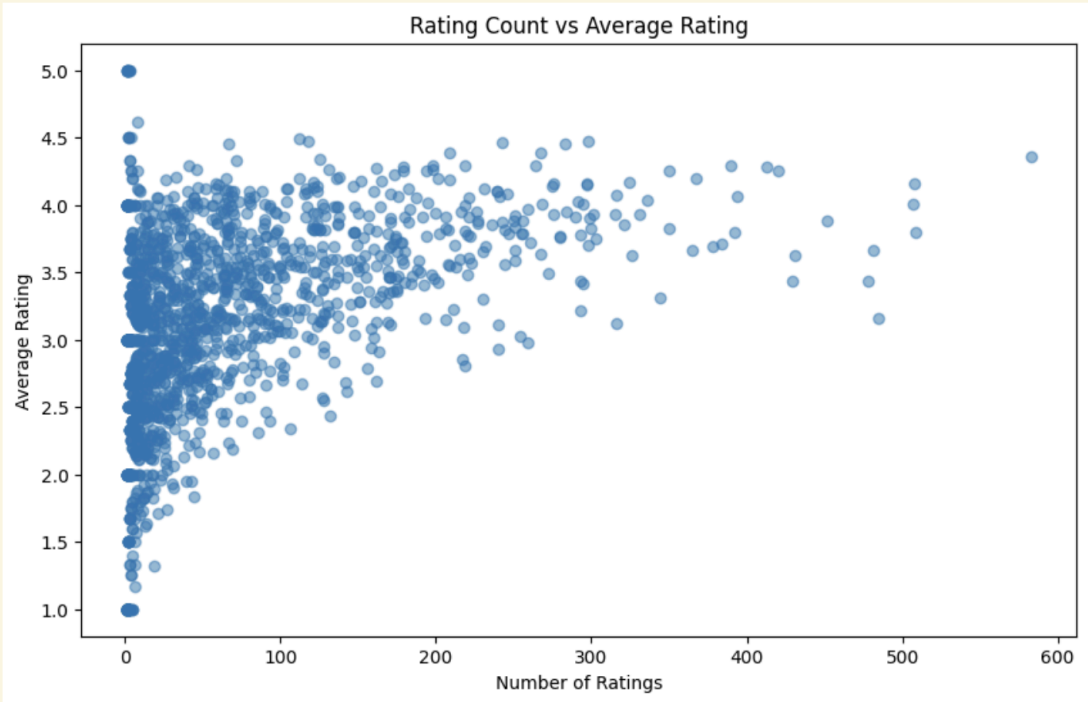
... ☒ Rating trends analysis:

Top 10 most rated movies:

	title	release_date	rating_count	avg_rating	\
0	Star Wars (1977)	1977-01-01	583	4.36	
1	Contact (1997)	1997-07-11	509	3.80	
2	Fargo (1996)	1997-02-14	508	4.16	
3	Return of the Jedi (1983)	1997-03-14	507	4.01	
4	Liar Liar (1997)	1997-03-21	485	3.16	
5	English Patient, The (1996)	1996-11-15	481	3.66	
6	Scream (1996)	1996-12-20	478	3.44	
7	Toy Story (1995)	1995-01-01	452	3.88	
8	Air Force One (1997)	1997-01-01	431	3.63	
9	Independence Day (ID4) (1996)	1996-07-03	429	3.44	

unique_raters

0	583
1	509
2	508
3	507
4	485
5	481
6	478
7	452
8	431
9	429



Answering our business questions

1. Which movies are the most popular among different age groups?

- Movies vary among age groups. Younger demographics like students prefer Comedy whereas retired individuals favor Drama.
- Copycat was the highest rated movie in general.

2. Do male and female users have different rating preferences?

- Females tend to prefer musicals, children movies and romance movies.
- Males tend to prefer westerns, mysteries, and sci-fi movies.
- The largest gender gap appears in musicals with a +0.17 difference leaning towards females.

3. Which movie genres have the highest average ratings?

- Crime genre had the highest average ratings.
- Drama genre has the highest number of movies, followed by Comedy and Action, while genres like Film-Noir and Fantasy have fewer movies.

4. How are users' ratings affected by movie release date?

- Average ratings show a gradual decline from the 1940s to 2000s, with older films generally rated higher.
- Rating volume explodes from the 1980s forward, with the most ratings occurring for 1990s films.

5. How does movie rating behavior vary by user occupation?

- Average ratings remain consistent across occupations (3.3-3.8)
- Genre preferences show subtle variations by professional background
- **Tech & Engineering:** Higher Comedy preference; Programmers show lower Drama interest
- **Business:** Executives favor Comedy; Marketing prefers Drama; Administrative staff lean toward Romance
- **Creative:** Writers and Artists show stronger Drama preferences
- **Public Service:** Educators and Healthcare professionals have highest Drama preferences
- **Age-Related:** Students prefer Comedy; Retired individuals favor Drama
- **Key Insight:** Occupational differences exist but are subtle - all professions maintain interest across all genres with only modest variations
- **Pattern:** Professional focus sometimes aligns with genre preference (Writers→Drama) but personal taste appears to transcend occupation

6. Which users are most active in terms of number of ratings?

- Students contribute the most total ratings
- Healthcare professionals average 175 ratings per user—the highest engagement rate